

Student Project Positions in Robot Learning

Several open positions in the Robotic Intelligence Lab at the University of Cambridge

The Robotic Intelligence Lab opens at the University of Cambridge in January 2027, and we are looking for several visiting students to join from the beginning, for Master's theses and other research projects. Projects are open to discussion and are defined together with each student, broadly within general-purpose robot manipulation, vision-language-action models, diffusion policies, imitation learning, reinforcement learning, robot collaboration and aerial manipulation. We welcome both Cambridge students and visiting students from other universities.

PROJECT	Thesis or student	STUDENT	Cambridge or visiting
SEMESTER	From January 2027	HOST	University of Cambridge
START	January 2027 or later	SUPERVISOR	Prof. Ioannis Mandralis
DEADLINE	Set by the student's programme	ADVISORS	Visiting students: co-advisor at home institution (found by the student)
CONTACT imm38@cam.ac.uk · mandralis.github.io/Robotic-Intelligence-Lab			

1 The lab

The Robotic Intelligence Lab is a new robotics group in the Machine Intelligence Group of the Division of Information Engineering, Department of Engineering. We aim to give robots the intelligence and agility they need to do useful physical work in the real world, combining robot learning, control theory and real hardware. Our first projects centre on robots that grasp, lift and reorient objects, alone and in teams, using only their own onboard sensing. Students who join in the first year help build the lab's first platforms in a stimulating research environment.

2 Research areas

Projects are open to discussion, and will be broadly in the following areas.

- **General-purpose robot manipulation.** Policies that grasp and manipulate a wide range of objects, including objects not seen in training, on single arms, bimanual stations and mobile platforms.
- **Vision-language-action models.** Fine-tuning pretrained models on new embodiments and

tasks, with goals given in plain language, such as the object to pick up.

- **Diffusion policies.** Expressive visuomotor policies learned from demonstrations, and how to make them account for the dynamics of the robot that executes them.
- **Imitation learning.** Learning from teleoperated and handheld demonstrations, and distilling privileged experts trained in simulation into policies that use onboard cameras.
- **Reinforcement learning.** Massively parallel training in simulation, transfer to hardware, reinforcement-learning fine-tuning of pretrained policies, and continual learning without forgetting.
- **Robot collaboration.** Several robots sharing one task, such as lifting, handing over or reorienting an object together, using multi-agent learning and tools from control theory for the coupled system.
- **Aerial manipulation.** Drones with grippers and arms that grasp, lift, carry and reorient objects using only onboard sensing.

Students are equally welcome to propose an idea of their own within these areas and we are happy to discuss.

3 Who can apply and how it works

- **Project types.** Master's theses, Cambridge Part IIB and MPhil projects, and semester projects or research internships for visiting students from other universities. Prof. Mandralis has had success in working with students from (ETHZ, EPFL, TUM, Caltech, etc.). **>6 month research stays are strongly preferred.**
- **Place and timing.** Projects are carried out in the lab in Cambridge, from January 2027 onwards. Start date and duration follow the rules of the student's programme.
- **Topic.** We agree the topic together before the start, matched to the student's interests and to the lab's directions, and aim where possible for results that can be published.
- **Co-supervision for visiting students.** Projects are supervised in Cambridge and co-advised by a supervisor at the student's home institution. **It is the student's responsibility to find this local departmental supervisor**, who can formally supervise and grade the project, and to register the project with the home institution. We are glad to talk with prospective co-advisors.
- **Funding.** Visiting students cover their own travel and living costs, for example through mobility grants or scholarships from their home institution.

4 Candidate profile

- An excellent student in robotics, mechanical or electrical engineering, computer science or a related field.
- Solid foundations in machine learning and/or control, and fluency in Python and PyTorch; experience with robot simulation (e.g. Isaac Lab, MuJoCo) or hardware (e.g. ROS 2) is a plus.
- Independence, and the drive to make methods work on real robots.

5 Applying

Send a CV, a transcript of records and a short paragraph on which of the areas above interest you most to imm38@cam.ac.uk, with "Student project" in the subject line. Please include the type of

project, your preferred start date and duration and, for visiting students, the name of your local co-advisor if you already have one in mind. Positions are filled on a rolling basis.